Technical Data Table 2 P.

IP55 IC411

Insulation class F
Temperature rise class B

Output kW	Type designation	380V 50Hz Speed n r/min	Efficiency η %	Power factor cos ϕ	Current A	415V 50Hz Speed n r/min	Effciency η %	Power factor cos ϕ	Current A	Moment of inertia J=GD²/4 kgm²	Weight kg	Sound pressure level Lp dB(A)
					3000r/min=2poles Basic design							
0.37 **M2QA**	71M2A	2765	71.0	0.85	0.93	2790	72.0	0.80	0.89	0.00030	10	56
0.55	71M2B	2780	73.5	0.85	1.34	2800	75.5	0.81	1.25	0.00037	11	56
0.75	80M2A	2825	73.7	0.86	1.80	2855	74.7	0.84	1.66	0.00091	16	57
1.1	80M2B	2840	76.2	0.87	2.52	2870	77.4	0.845	2.34	0.00107	17	58
1.5	90S2A	2835	77.8	0.885	3.31	2865	78.8	0.86	3.08	0.00135	21	61
2.2	90L2A	2835	80.2	0.88	4.74	2865	81.0	0.83	4.55	0.00163	24	61
3	100L2A	2845	82.0	0.88	6.32	2870	82.8	0.845	5.97	0.00402	33	65
4	112M2A	2860	83.1	0.915	7.99	2890	84.4	0.88	7.49	0.00671	42	67
5.5	132S2A	2890	84.8	0.90	10.9	2910	86.0	0.875	10.2	0.01241	58	70
7.5	132S2B	2900	86.0	0.90	14.7	2920	87.2	0.89	13.4	0.01491	63	70
11	160M2A	2910	87.6	0.885	21.6	2930	88.2	0.85	20.4	0.0436	112	72
15	160M2B	2910	89.3	0.895	28.5	2930	89.8	0.875	26.6	0.0551	122	72
18.5	160L2A	2910	89.8	0.905	34.6	2930	90.3	0.885	32.2	0.06549	142	72
22	180M2A	2935	90.4	0.91	40.6	2950	90.4	0.88	38.5	0.08805	170	75
30	200L2A	2950	91.2	0.91	54.9	2960	91.2	0.89	51.4	0.14821	235	81
37	200L2B	2950	91.6	0.915	67.1	2960	91.6	0.895	62.8	0.16822	254	81
45	225M2A	2965	92.1	0.90	82.5	2970	92.1	0.875	77.7	0.29345	328	81
55	250M2A	2960	92.3	0.905	100	2965	92.5	0.89	92.9	0.3784	400	84
75	280S2A	2965	92.8	0.92	133	2970	93.1	0.905	124	0.587	498	85
90	280M2A	2965	93.1	0.925	159	2970	93.4	0.915	147	0.615	554	85
110	315S2A	2975	93.6	0.905	197	2980	93.6	0.89	184	1.4083	869	88
132	315M2A	2975	94.2	0.905	235	2980	94.2	0.89	219	1.5584	962	88
160	315L2A	2975	94.2	0.91	284	2980	94.2	0.89	266	1.7256	1014	88
*200	315L2B	2970	94.4	0.915	352	2975	94.4	0.90	327	1.9405	1166	88
*250	355M2A	2980	94.5	0.905	444	2980	94.5	0.895	411	3.05	1658	89
*315	355L2A	2980	94.8	0.905	558	2980	94.8	0.895	517	3.6	1815	89

Output kW	Type designation	380V 50Hz Speed n r/min	Efficiency η %	Power factor cos ϕ	Current A	415V 50Hz Speed n r/min	Effciency η %	Power factor cos ϕ	Current A	Moment of inertia J=GD²/4 kgm²	Weight kg	Sound pressure level Lp dB(A)
4 P.					1500r/min=4poles Basic design							
0.25 **M2QA**	71M4A	1385	68.0	0.73	0.77	1405	69.0	0.67	0.75	0.00053	11	43
0.37	71M4B	1385	68.0	0.745	1.11	1405	69.0	0.69	1.08	0.00066	11	45
0.55	80M4A	1400	71.5	0.75	1.56	1420	72.0	0.70	1.52	0.00145	16	46
0.75	80M4B	1405	73.2	0.78	2.00	1425	74.7	0.735	1.90	0.00174	17	46
1.1	90S4A	1385	75.7	0.785	2.81	1405	76.5	0.735	2.72	0.00254	21	52
1.5	90L4A	1390	77.8	0.795	3.68	1410	78.8	0.755	3.51	0.00317	25	52
2.2	100L4A	1420	80.2	0.81	5.15	1435	81.0	0.765	4.94	0.00679	33	53
3	100L4B	1415	81.7	0.83	6.72	1430	82.8	0.785	6.42	0.00862	36	53
4	112M4A	1430	83.6	0.805	9.03	1440	84.1	0.75	8.82	0.01306	45	56
5.5	132S4A	1425	84.7	0.84	11.7	1440	85.6	0.79	11.3	0.02673	60	59
7.5	132M4A	1430	86.1	0.84	15.8	1450	87.1	0.80	15.0	0.03432	73	59
11	160M4A	1450	88.2	0.87	21.8	1460	88.2	0.83	20.9	0.06543	116	66
15	160L4A	1450	89.3	0.88	29.0	1460	89.3	0.85	27.5	0.09349	137	66
18.5	180M4A	1465	89.8	0.88	35.6	1470	89.8	0.85	33.7	0.16049	170	66
22	180L4A	1465	90.4	0.89	41.5	1470	90.4	0.86	39.4	0.18046	186	66
30	200L4A	1470	91.2	0.885	56.5	1475	91.2	0.855	53.5	0.2819	254	71
37	225S4A	1475	91.6	0.875	70.1	1480	91.6	0.84	66.9	0.37	308	73
45	225M4A	1475	92.1	0.865	85.8	1480	92.1	0.835	81.4	0.42	335	73
55	250M4A	1475	92.3	0.89	102	1480	92.6	0.87	95.0	0.78	433	76
75	280S4A	1480	92.9	0.89	138	1480	93.1	0.875	128	1.10	510	78
90	280M4A	1480	93.4	0.90	163	1485	93.6	0.885	151	1.35	583	78
110	315S4A	1480	93.6	0.875	204	1485	93.6	0.855	191	2.8596	904	80
132	315M4A	1480	93.8	0.885	242	1485	93.8	0.865	226	3.1848	1003	80
160	315L4A	1480	94.0	0.885	292	1485	94.0	0.86	275	3.6765	1095	86
200	315L4B	1480	94.2	0.885	364	1485	94.2	0.865	341	4.2516	1204	86
*250	355M4A	1485	94.3	0.905	445	1490	94.4	0.895	412	6.77	1668	87
*315	355L4A	1485	94.7	0.905	558	1490	94.8	0.895	517	8.2	1871	87

Code letters For supplementing the product code
Code letters For voltage and-frequency(product code position 13)

*Insulation Class F Temperature rise Class F

D	S	
380~420V△50Hz	220~240V△50Hz	Other rated voltage connection or frequency(Max,690V)can be used with VC002 or VC209.(The meaning of Varient code can be referred to Variant Code List on Page 25)
660~690VY50Hz	380~420VY50Hz	
440~480[1)]V△60Hz	440~480VY60Hz	

[1)]480V not stamped on sizes 160 to 355

Efficiency values are giving according to IEC 60034-2-1:2007
Please note that the values are not comparable without knowing the testing method.
ABB has calculated the efficiency values according to indirect method,stray load losses(additional losses)determined from measuring.

IE-class concerns motors from 0.75kW to 375kW and 2,4,6 poles.

Technical Data Table 6 P.

IP55 IC411

Insulation class F
Temperature rise class B

Output kW	Type designation		Product code	Speed n r/min	Efficiency Full load 100% η	Efficiency 3/4 load 75% η	Power factor cos ϕ	Current IN A	Current Is/IN	Torque TN Nm	Torque Ts/TN	Torque TMAX/TN
400V 50Hz												
1000r/min=6poles Basic design												
0.18	M2QA 71M6A	3GQA	073301-**A	885	55.0	54.0	0.67	0.71	3.5	1.94	2.1	2.3
0.25	71M6B		073302-**A	885	58.0	59.6	0.67	0.93	3.5	2.70	2.1	2.3
0.37	80M6A		083301-**A	930	63.5	63.9	0.66	1.27	5.0	3.80	1.9	1.8
0.55	80M6B		083302-**A	925	65.7	66.9	0.675	1.79	5.0	5.68	1.9	1.8
0.75	90S6A		093101-**A	920	71.5	72.6	0.72	2.10	5.0	7.79	2.0	2.2
1.1	90L6A		093501-**A	920	74.3	75.5	0.74	2.89	5.0	11.4	2.0	2.2
1.5	100L6A		103501-**A	940	76.4	76.2	0.74	3.83	5.5	15.2	2.0	2.2
2.2	112M6A		113301-**A	940	78.8	79.8	0.73	5.52	5.5	22.4	2.0	2.2
3	132S6A		133101-**A	945	80.7	81.6	0.77	6.97	6.5	30.3	2.0	2.2
4	132M6A		133301-**A	950	82.3	82.9	0.77	9.11	6.5	40.2	2.0	2.2
5.5	132M6B		133302-**A	950	83.9	85.8	0.78	12.1	6.5	55.3	2.0	2.2
7.5	160M6A		163301-**A	960	85.5	86.4	0.78	16.2	6.0	74.6	2.0	2.3
11	160L6A		163501-**A	970	87.1	88.0	0.78	23.4	6.0	108	2.2	2.3
15	180L6A		183501-**A	975	88.3	88.7	0.82	29.9	6.0	147	2.3	2.8
18.5	200L6A		203501-**A	980	89.2	90.2	0.82	36.5	6.0	180	2.2	2.8
22	200L6B		203502-**A	980	89.7	90.4	0.83	42.7	6.0	214	2.1	2.8
30	225M6A		223301-**A	985	90.7	91.1	0.815	58.6	6.6	291	2.2	2.8
37	250M6A		253301-**A	975	91.3	91.6	0.87	67.2	6.8	362	2.3	2.8
45	280S6A		283101-**A	985	91.8	92.0	0.875	80.9	6.2	436	2.3	2.4
55	280M6A		283301-**A	985	92.3	92.4	0.875	98.3	7.0	533	2.3	2.5
75	315S6A		313101-**A	985	93.0	93.2	0.86	135	7.4	727	2.0	2.0
90	315M6A		313301-**A	985	93.3	93.5	0.86	162	7.4	872	2.0	2.0
110	315L6A		313501-**A	985	93.6	93.8	0.875	194	6.8	1066	2.0	2.0
132	315L6B		313502-**A	985	93.8	94.0	0.875	232	6.8	1280	2.0	2.0
*160	355M6A		353301-**A	990	94.0	94.2	0.88	279	6.8	1543	2.1	2.4
*200	355M6B		353302-**A	990	94.2	94.3	0.88	348	6.7	1929	2.0	2.3
*250	355L6A		353501-**A	990	94.4	94.5	0.88	434	6.7	2412	2.0	2.3
400V 50Hz												
8 P.												
750r/min=8poles Basic design												
0.18	M2QA 80M8A	3GQA	084301-**A	700	51.0	50.1	0.60	0.85	3.3	2.46	1.8	1.9
0.25	80M8B		084302-**A	700	54.5	53.2	0.60	1.10	3.6	3.41	1.8	1.9
0.37	90S8A		094101-**A	700	62.5	62.1	0.605	1.41	4.4	5.05	1.8	1.9
0.55	90L8A		094501-**A	700	63.5	63.3	0.615	2.03	4.7	7.50	1.8	2.0
0.75	100L8A		104501-**A	690	68.5	68.6	0.64	2.47	5.0	10.4	1.8	2.0
1.1	100L8B		104502-**A	675	71.3	70.1	0.645	3.45	5.0	15.6	1.8	2.0
1.5	112M8A		114301-**A	695	74.2	74.6	0.675	4.32	5.0	20.6	1.8	2.0
2.2	132S8A		134101-**A	710	79.8	80.6	0.70	5.68	5.5	29.6	1.8	2.0
3	132M8A		134301-**A	710	80.0	80.4	0.75	7.22	5.5	40.4	1.8	2.0
4	160M8A		164301-**A	720	83.0	83.0	0.73	9.53	5.5	53.1	2.1	2.5
5.5	160M8B		164302-**A	720	84.5	94.6	0.74	12.7	5.5	73.0	2.1	2.5
7.5	160L8A		164501-**A	720	85.2	84.5	0.74	17.2	5.5	99.5	2.1	2.5
11	180L8A		184501-**A	730	87.5	86.8	0.77	23.6	5.4	144	2.0	2.8
15	200L8A		204501-**A	730	89.0	89.4	0.775	31.4	5.5	196	2.3	2.8
18.5	225S8A		224101-**A	735	89.5	88.6	0.73	40.9	5.5	240	2.1	2.8
22	225M8A		224301-**A	735	89.7	88.8	0.74	47.8	6.0	286	2.2	2.8
30	250M8A		254301-**A	730	91.3	89.3	0.79	60.0	6.5	392	2.3	2.6
37	280S8A		284101-**A	735	91.2	91.0	0.80	73.2	6.0	481	2.1	2.6
45	280M8A		284301-**A	735	92.0	90.6	0.80	88.2	6.0	585	2.1	2.7
55	315S8A		314101-**A	740	92.5	91.2	0.82	105	6.9	710	1.8	2.0
75	315M8A		314301-**A	740	93.0	91.9	0.82	142	7.0	968	1.8	2.0
90	315L8A		314501-**A	740	93.5	92.9	0.82	169	7.1	1161	1.8	2.0
110	315L8B		314502-**A	740	94.0	92.4	0.825	205	6.4	1420	1.8	2.0
*132	355M8A		354301-**A	740	94.2	93.9	0.815	248	6.2	1704	1.8	2.0
*160	355M8B		354302-**A	740	94.5	94.1	0.82	298	6.2	2065	1.8	2.0
*200	355L8A		354501-**A	740	94.7	94.5	0.82	372	6.2	2581	1.8	2.0

Code letters For supplementing the product code
Code letters For voltage and-frequency(product code position 13)

*Insulation Class F Temperature rise Class F

D	S
380~420V△50Hz	220~240V△50Hz
660~690VY50Hz	380~420VY50Hz
440~480[1)]V△60Hz	440~480VY60Hz

Other rated voltage connection or frequency(Max,690V)can be used with VC002 or VC209.(The meaning of Varient code can be referred to Variant Code List on Page 25)

[1)]480V not stamped on sizes 160 to 355

Efficiency values are giving according to IEC 60034-2-1:2007
Please note that the values are not comparable without knowing the testing method.
ABB has calculated the efficiency values according to indirect method,stray load losses(additional losses)determined from measuring.

IE-class concerns motors from 0.75kW to 375kW and 2,4,6 poles.

Technical Data Table

IP55 IC411

Insulation class F
Temperature rise class B

		380V 50Hz				415V 50Hz						
Output kW	Type designation	Speed n r/min	Efficiency η %	Power factor cos ϕ	Current A	Speed n r/min	Effciency η %	Power factor cos ϕ	Current A	Moment of inertia $J=GD^2/4$ kgm^2	Weight kg	Sound pressure level Lp dB(A)
1000r/min=6poles Basic design												
0.18 **M2QA**	71M6A	880	52.0	0.69	0.76	890	55.0	0.63	0.72	0.00056	10	42
0.25	71M6B	880	56.0	0.69	0.98	890	58.0	0.66	0.91	0.00074	11	42
0.37	80M6A	925	63.0	0.685	1.30	935	63.0	0.63	1.30	0.00159	17	45
0.55	80M6B	920	65.0	0.705	1.82	930	65.5	0.635	1.84	0.00196	18	45
0.75	90S6A	915	70.5	0.755	2.14	930	71.7	0.70	2.08	0.00292	21	48
1.1	90L6A	915	73.3	0.77	2.96	925	74.5	0.72	2.85	0.00379	25	48
1.5	100L6A	930	75.6	0.76	3.97	945	76.8	0.715	3.80	0.00999	32	51
2.2	112M6A	935	78.2	0.75	5.70	945	79.2	0.71	5.44	0.01559	40	54
3	132S6A	940	80.1	0.78	7.30	950	81.1	0.75	6.86	0.03116	55	56
4	132M6A	945	81.4	0.78	9.57	955	82.8	0.75	8.96	0.04074	65	56
5.5	132M6B	940	83.3	0.79	12.7	955	84.3	0.765	11.9	0.05332	75	56
7.5	160M6A	950	85.5	0.79	16.9	970	85.5	0.75	16.3	0.09231	119	61
11	160L6A	965	87.1	0.80	24.0	975	87.1	0.76	23.1	0.12970	140	62
15	180L6A	975	88.3	0.84	30.7	980	88.3	0.80	29.5	0.2418	180	63
18.5	200L6A	975	89.2	0.84	37.5	980	89.2	0.80	36.1	0.34174	231	64
22	200L6B	975	89.7	0.84	44.4	980	89.7	0.81	42.1	0.46837	254	64
30	225M6A	980	90.7	0.835	60.2	985	90.7	0.79	58.2	0.62691	308	66
37	250M6A	975	90.9	0.88	70.3	980	91.5	0.86	65.4	0.97	385	68
45	280S6A	980	91.6	0.885	84.3	985	91.8	0.865	78.8	1.25	467	69
55	280M6A	980	92.0	0.885	103	985	92.3	0.865	95.8	1.485	520	70
75	315S6A	985	92.8	0.87	141	985	93.0	0.85	132	3.1942	809	70
90	315M6A	985	93.2	0.87	169	985	93.3	0.85	158	3.723	918	70
110	315L6A	985	93.5	0.88	203	985	93.6	0.87	188	4.2564	1003	70
132	315L6B	985	93.6	0.88	243	985	93.8	0.87	225	5.1577	1076	70
*160	355M6A	990	94.0	0.89	291	990	94.0	0.86	275	7.8	1605	75
*200	355M6B	990	94.1	0.895	361	990	94.2	0.875	338	9.1	1778	75
*250	355L6A	990	94.3	0.895	450	990	94.4	0.87	423	11.4	1990	75

		380V 50Hz				415V 50Hz						
Output kW	Type designation	Speed n r/min	Efficiency η %	Power factor cos ϕ	Current A	Speed n r/min	Effciency η %	Power factor cos ϕ	Current A	Moment of inertia $J=GD^2/4$ kgm^2	Weight kg	Sound pressure level Lp dB(A)
750r/min=8poles Basic design												
0.18 **M2QA**	80M8A	695	48.0	0.63	0.90	705	52.5	0.57	0.84	0.00111	16	42
0.25	80M8B	695	52.3	0.63	1.15	705	55.0	0.57	1.11	0.00326	17	42
0.37	90S8A	695	62.0	0.62	1.46	705	62.5	0.59	1.40	0.00541	21	46
0.55	90L8A	695	63.0	0.63	2.11	705	63.5	0.60	2.01	0.00756	24	46
0.75	100L8A	680	67.0	0.65	2.62	695	69.0	0.62	2.44	0.00971	31	53
1.1	100L8B	665	68.8	0.68	3.57	685	71.8	0.625	3.41	0.01186	34	53
1.5	112M8A	690	73.2	0.68	4.58	700	74.4	0.64	4.38	0.01559	42	55
2.2	132S8A	705	79.3	0.745	5.66	715	79.8	0.685	5.60	0.03625	56	55
3	132M8A	705	79.5	0.78	7.35	715	80.0	0.725	7.20	0.04141	64	56
4	160M8A	715	83.0	0.76	9.63	725	83.0	0.70	9.58	0.0676	105	58
5.5	160M8B	715	84.5	0.76	13.0	720	84.5	0.72	12.6	0.09524	125	58
7.5	160L8A	715	85.2	0.77	17.4	720	85.2	0.72	17.0	0.12122	142	58
11	180L8A	725	87.5	0.79	24.2	730	87.5	0.75	23.3	0.23645	176	62
15	200L8B	725	89.0	0.80	32.0	730	89.0	0.75	31.3	0.37103	235	63
18.5	225S8A	735	89.5	0.75	41.9	740	89.3	0.69	41.8	0.53287	290	65
22	225M8A	735	89.7	0.76	49.0	740	89.5	0.70	48.9	0.65825	302	65
30	250M8A	730	91.2	0.81	61.7	735	91.4	0.77	59.3	0.975	390	67
37	280S8A	735	91.1	0.81	76.2	735	91.2	0.785	71.9	1.25	471	68
45	280M8A	735	91.9	0.81	91.8	735	92.0	0.79	86.1	1.485	537	68
55	315S8A	735	92.3	0.83	109	740	92.6	0.805	103	3.6842	779	65
75	315M8A	735	92.8	0.83	148	740	93.1	0.805	139	4.9591	941	68
90	315L8A	740	93.3	0.83	177	740	93.6	0.805	166	5.8205	1022	68
110	315L8B	735	93.8	0.84	212	740	94.1	0.81	201	6.7537	1104	68
*132	355M8A	740	94.1	0.83	257	740	94.3	0.80	243	8.6	1585	71
*160	355M8B	740	94.4	0.835	308	740	94.6	0.805	292	10.01	1740	71
*200	355L8A	740	94.6	0.835	385	740	94.8	0.805	365	12.5	1946	71

Code letters For supplementing the product code
Code letters For voltage and-frequency(product code position 13)

*Insulation Class F Temperature rise Class F

D	S
380~420V△50Hz	220~240V△50Hz
660~690VY50Hz	380~420VY50Hz
440~480[1)]V△60Hz	440~480VY60Hz

Other rated voltage connection or frequency(Max,690V)can be used with VC002 or VC209.(The meaning of Varient code can be referred to Variant Code List on Page 25)

[1)]480V not stamped on sizes 160 to 355

Efficiency values are giving according to IEC 60034-2-1:2007
Please note that the values are not comparable without knowing the testing method.
ABB has calculated the efficiency values according to indirect method,stray load losses(additional losses)determined from measuring.

IE-class concerns motors from 0.75kW to 375kW and 2,4,6 poles.